



Mathematics: analysis and approaches
Higher level
Paper 3

20 May 2026

Zone A afternoon | Zone B afternoon | Zone C afternoon

1 hour 15 minutes

Instructions to students

- Do not open this examination paper until instructed to do so.
- A graphic display calculator is required for this paper.
- Answer all the questions in the answer booklet provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches HL formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[55 marks]**.

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Answer **all** questions in the answer booklet provided. Please start each question on a new page. Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Solutions found from a graphic display calculator should be supported by suitable working. For example, if graphs are used to find a solution, you should sketch these as part of your answer. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

1. [Maximum mark: 30]

This question considers different methods of summing powers of integers.

The sum of the first n positive integers, each raised to the power zero, is given by

$$S_0(n) = 1^0 + 2^0 + \dots + n^0 \text{ where } n \in \mathbb{Z}^+.$$

(a) (i) Find the value of $S_0(4)$. [1]

(ii) Write down a simplified expression for $S_0(n)$. [1]

The sum of the first n positive integers, each raised to the power 1, is given by

$$S_1(n) = 1^1 + 2^1 + \dots + n^1 \text{ where } n \in \mathbb{Z}^+.$$

(b) Use the fact that $S_1(n)$ is the sum of n terms of an arithmetic sequence to show

that $S_1(n) = \frac{n^2}{2} + \frac{n}{2}$. [3]

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(Question 1 continued)

The sum of the first n square numbers is given by

$$S_2(n) = 1^2 + 2^2 + 3^2 \dots + n^2 \text{ where } n \in \mathbb{Z}^+.$$

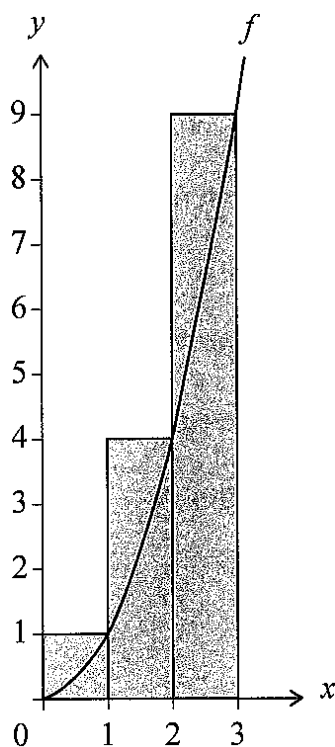
(c) Find $S_2(5)$.

[1]

(d) The following graph shows three rectangles, where the heights of the rectangles are consecutive square numbers, and the width of each rectangle is 1.

The graph may be extended in a similar way to include any number of rectangles.

The curve $y = f(x)$ passes through the top right corner of each rectangle.



(i) Write down an expression for $f(x)$.

[1]

(ii) By considering the total area of the first n rectangles and the area under the curve, show that $S_2(n) > \frac{n^3}{3}$.

[4]

(iii) Use a similar method to show that $S_2(n) < \frac{n^3}{3} + n^2 + n$.

[4]

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(Question 1 continued)

Consider the sequence $u_n = n^3 - (n-1)^3$ for $n \in \mathbb{Z}^+$. Some of the values of u_n and $\sum_{r=1}^n u_r$ are shown in the following table.

n	u_n	$\sum_{r=1}^n u_r$
1	1	1
2	7	8
3	a	b
4	37	64

- (e) Find the value of
- (i) a ; [1] 311
- (ii) b . [1]
- (f) (i) Suggest an expression for $\sum_{r=1}^n u_r$ in terms of n . [1]
- (ii) Use the definition of u_n to prove that your suggestion is true. [3]
- (g) (i) Find a quadratic expression for u_n . [2]
- (ii) Hence, write $\sum_{r=1}^n u_r$ in terms of $S_2(n)$, $S_1(n)$ and $S_0(n)$. [2]
- (iii) Hence, show that $S_2(n) = \frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}$. [3] A000
- (iv) Show that the expression for $S_2(n)$ given in (g)(iii) is consistent with the inequalities in part (d). [2]



$$(a) (i) S_0(4) = 1^0 + 2^0 + 3^0 + 4^0 = 4$$

$$(ii) S_0(n) = n$$

$$(b) S_1(n) = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2}$$

$$(c) S_2(5) = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$$

$$(d) (i) f(x) = x^2$$

$$(ii) S_2(n) > \int_0^n f(x) dx = \int_0^n x^2 dx = \frac{1}{3} [x^3]_0^n = \frac{1}{3} n^3 - 0 = \frac{1}{3} n^3$$

$$(iii) S_2(n) < \int_1^{n+1} f(x) dx = \int_1^{n+1} x^2 dx = \frac{1}{3} [x^3]_1^{n+1} = \frac{1}{3} (n+1)^3 - \frac{1}{3} = \frac{1}{3} n^3 + n^2 + n$$

$$(e) (i) a = u_3 = 3^3 - 2^3 = 27 - 8 = 19$$

$$(ii) b = 8 + a = 8 + 19 = 27$$

$$(f) (i) \sum_{r=1}^n u_r = n^3$$

$$(ii) u_n = n^3 - (n-1)^3$$

$$u_{n-1} = (n-1)^3 - (n-2)^3$$

$$\vdots$$

$$u_2 = 2^3 - 1^3$$

$$u_1 = 1^3 - 0^3$$

Adding together: $u_1 + u_2 + \dots + u_n = n^3 - 0^3$

$$\Rightarrow \sum_{r=1}^n u_r = n^3.$$

$$\begin{aligned} (7) \text{ (i)} \quad u_n &= n^3 - (n-1)^3 \\ &= n^3 - (n^3 - 3n^2 + 3n - 1) \\ &= 3n^2 - 3n + 1 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \sum_{r=1}^n u_r &= \sum_{r=1}^n (3r^2 - 3r + 1) \\ &= 3 \sum_{r=1}^n r^2 - 3 \sum_{r=1}^n r + n \\ &= 3S_2(n) - 3S_1(n) + S_0(n) \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad n^3 &= 3S_2(n) - 3 \times \frac{n(n+1)}{2} + n \\ \Rightarrow 3S_2(n) &= n^3 + \frac{3n^2 + 3n}{2} - n \\ &= \frac{2n^3 + 3n^2 + 3n - 2n}{2} \\ &= n^3 + \frac{3}{2}n^2 + n \end{aligned}$$

$$\Rightarrow S_2(n) = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n$$

(iv) from $f(iii)$: $S_2(n) = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n$

$$S_2(n) - \frac{n^3}{3} = \left(\frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n\right) - \frac{n^3}{3} = \frac{1}{2}n^2 + \frac{1}{3}n > 0$$

$\Rightarrow S_2(n) > \frac{n^3}{3}$ Which is consistent with the result in part d(ii)

$$S_2(n) - \left(\frac{n^3}{3} + n^2 + n\right) = \left(\frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{3}n\right) - \left(\frac{n^3}{3} + n^2 + n\right) = -\frac{1}{2}n^2 - \frac{2}{3}n < 0$$

$\Rightarrow S_2(n) < \frac{n^3}{3} + n^2 + n$ Which is consistent with the result in part d(iii)

In conclusion: The expression for $S_2(n)$ given in $f(iii)$ is consistent with the inequalities in part (d).

2. [Maximum mark: 25]

This question asks you to explore curves in the complex plane defined by $r = a + b \sin \theta$, where a and b are real constants.

A complex number z can be represented by a point in the complex plane whose position is determined by the modulus, r , and the argument, θ , of z .

The set of points which satisfy a given relationship between r and θ form a curve in the complex plane.

Consider a family of curves defined by $r = a + b \sin \theta$, where a and b are real constants.

First consider the case where $a = 4$ and $b = 0$ so that $r = 4$.

The curve defined by $r = 4$ is the set of points in the complex plane which represent complex numbers with a modulus of 4.

(a) Suppose that z is a complex number such that $r = 4$.

Find all possible values of z such that

- (i) z is purely real; [1]
- (ii) z is purely imaginary; [1]
- (iii) the argument of z is $\frac{\pi}{4}$. [1]

(b) Sketch the curve defined by $r = 4$ on an Argand diagram, showing clearly the values of the intercepts with the real and imaginary axes. [2]

Now consider the case where $a = 0$ and $b = 4$ so that $r = 4 \sin \theta$, where $0 \leq \theta \leq \pi$.

(c) Suppose that z is a complex number such that $r = 4 \sin \theta$, where $0 \leq \theta \leq \pi$.

- (i) Write down the greatest possible value of r . [1]
- (ii) For this value of r , find z . [2]

(d) Use the identities $x = r \cos \theta$ and $y = r \sin \theta$ to show that the points on the curve defined by $r = 4 \sin \theta$ lie on the curve $x^2 + (y - 2)^2 = 4$, where $x = \operatorname{Re}(z)$ and $y = \operatorname{Im}(z)$. [4]

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(Question 2 continued)

(e) Consider the curve $x^2 + (y - 2)^2 = 4$.

(i) Find an expression for $\frac{dy}{dx}$ in terms of x and y . [3]

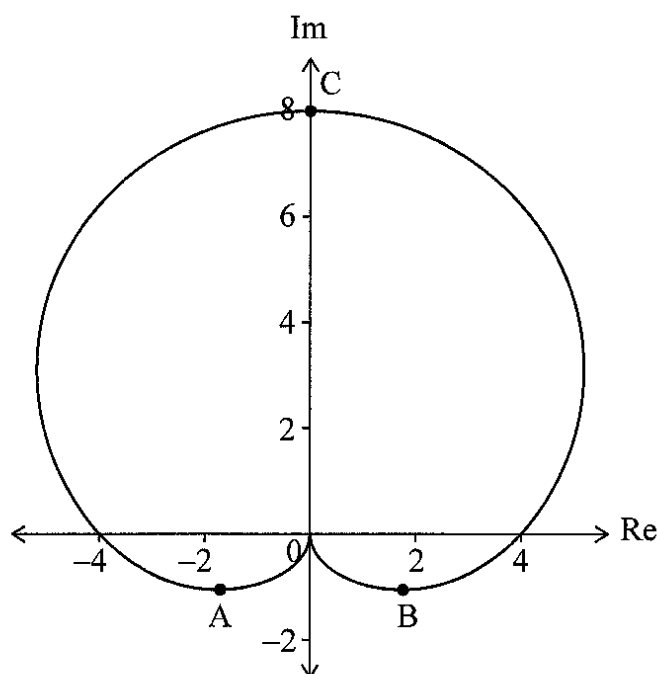
(ii) Hence or otherwise, find the values of x for which the gradient of the curve is undefined. [2]

(f) The curve defined by $r = 4 \sin \theta$, where $0 \leq \theta \leq \pi$, is a circle.

Sketch this curve on an Argand diagram, showing clearly the values of any intercepts with the real and imaginary axes. [1]

Finally, consider the case where $a = 4$ and $b = 4$.

The curve defined by $r = 4 + 4 \sin \theta$, where $0 \leq \theta \leq 2\pi$, is shown on the following Argand diagram. The curve has a horizontal tangent at point A, point B and point C. The real parts of the numbers represented by A, B and C are negative, positive and zero, respectively.



(g) Use the identities $x \equiv r \cos \theta$ and $y \equiv r \sin \theta$ to show that the gradient of the curve at the point $(r \cos \theta, r \sin \theta)$ is given by $\frac{dy}{dx} = \frac{\cos \theta (1 + 2 \sin \theta)}{1 - \sin \theta - 2 \sin^2 \theta}$. [5]

(h) Hence, find the value of θ which corresponds to the point A. [2]

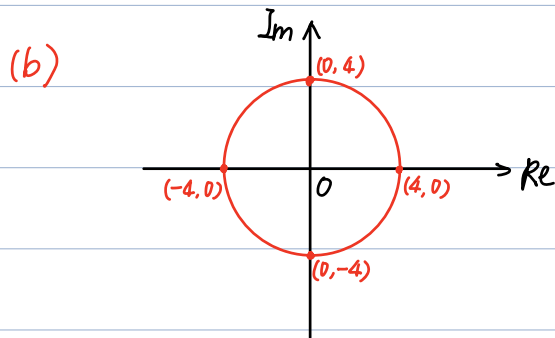


(a) $r = 4$

(i) $z = \pm 4$

(ii) $z = \pm 4i$

(iii) $z = 4 \operatorname{cis} \frac{\pi}{4} = 4 \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = 4 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} i \right) = 2\sqrt{2} + 2\sqrt{2}i$



(c) $r = 4 \sin \theta$

(i) $r_{\max} = 4$

(ii) $r_{\max} = 4$ when $\sin \theta = 1$ $\left. \begin{array}{l} 0 \leq \theta \leq \pi \end{array} \right\} \Rightarrow \theta = \frac{\pi}{2}$

$\therefore z = 4 \operatorname{cis} \frac{\pi}{2} = 4i$

(d) $r = 4 \sin \theta \Rightarrow r^2 = 4r \sin \theta \Rightarrow x^2 + y^2 = 4y \Rightarrow x^2 + y^2 - 4y = 0$

$\Rightarrow x^2 + (y^2 - 4y + 4) = 4 \Rightarrow x^2 + (y - 2)^2 = 4$

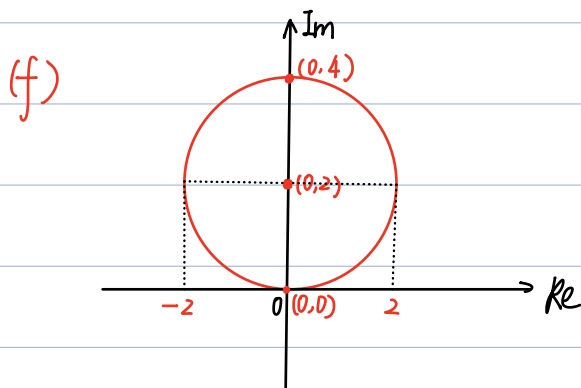
$$(e) (i) \quad 2x + 2(y-2) \frac{dy}{dx} = 0$$

$$\Rightarrow (y-2) \frac{dy}{dx} = -x$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y-2}$$

$$(ii) \quad \frac{dy}{dx} \text{ is undefined} \Rightarrow y-2=0 \Rightarrow y=2$$

$$\text{When } y=2 \Rightarrow x^2=4 \Rightarrow x=\pm 2$$



$$(g) \quad r = 4 + 4 \sin \theta \Rightarrow \frac{dr}{d\theta} = 4 \cos \theta$$

$$x = r \cos \theta \Rightarrow \frac{dx}{d\theta} = \frac{dr}{d\theta} \cos \theta + r \cdot (-\sin \theta) = 4 \cos \theta \cdot \cos \theta - (4 + 4 \sin \theta) \sin \theta$$

$$= 4 \cos^2 \theta - 4 \sin \theta - 4 \sin^2 \theta = 4(1 - \sin^2 \theta) - 4 \sin \theta - 4 \sin^2 \theta$$

$$= 4 - 4 \sin \theta - 8 \sin^2 \theta = 4(1 - \sin \theta - 2 \sin^2 \theta)$$

$$y = r \sin \theta \Rightarrow \frac{dy}{d\theta} = \frac{dr}{d\theta} \cdot \sin \theta + r \cdot \cos \theta = 4 \cos \theta \sin \theta + (4 + 4 \sin \theta) \cos \theta$$

$$= 4 \cos \theta + 8 \sin \theta \cos \theta = 4 \cos \theta (1 + 2 \sin \theta)$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} = \frac{dy}{d\theta} \cdot \frac{1}{\frac{dx}{d\theta}} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{4 \cos \theta (1 + 2 \sin \theta)}{4(1 - \sin \theta - 2 \sin^2 \theta)} = \frac{\cos \theta (1 + 2 \sin \theta)}{1 - \sin \theta - 2 \sin^2 \theta}$$

$$(h) \frac{dy}{dx} = 0 \Rightarrow \cos \theta (1 + 2 \sin \theta) = 0$$

$$\Rightarrow \cos \theta = 0 \quad \text{or} \quad \sin \theta = -\frac{1}{2}$$

$$0 \leq \theta < 2\pi$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

(rejected as $1 - \sin \theta - 2 \sin^2 \theta = 0$)

$$\therefore \theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$\therefore$ The real part of A is negative.

$\therefore$ A is in the 3rd Quadrant

$$\therefore \pi < \theta < \frac{3\pi}{2}$$

$$\therefore \theta = \frac{7\pi}{6} \text{ corresponds to the point A.}$$

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